AEROSENSE · AUDIO CORE

Technical Deep Dive & Working-Session Agenda

The H1 audio stack — architecture, hardware and software to build it out

Field	Detail
For	John Murray (audio / DSP) — pre-read + running order. In the room: John, Chris, Peter, Duran.
Session	Audio technical working session — Wednesday 15 July 2026, office
Scope	Audio only: Noise Cancelling, DVNC™, PureVoice™, VGSA™, DSMA™ audio delivery, and the audio-arbitration layer — plus the hardware and software each needs.
Basis	The per-technology DEV docs (Drive · Technologies/) + Flight Plan gates + Preflight 2026-07-08. This doc ties them together; those stay the source of truth.
Note	Internal / confidential — contains unfiled IP (DVNC). Keep external disclosure high-level until filed.

How to use this document

It is both the pre-read and the agenda. Sections 3.1–3.6 are the six working blocks for the day, in order. Each carries the what/why/how, the **hardware and software required to build it**, where it sits against the roadmap, and a short **"to decide on the day"** list. Sections 4–5 are the consolidated hardware bill and software stack; §7 is the timed agenda.

1. The audio architecture at a glance

One idea shapes everything: the AeroNode — a crash-protected processing + power hub, hard-mounted to the airframe — decouples heavy computation from the headset. That is what lets the audio stack do things a standalone headset cannot (a clean airframe vibration reference for DVNC; a high-rate attitude source for VGSA/DSMA). The H1 earcup runs only the hard-real-time inner loops; the AeroNode does the smart, non-real-time work.

1.1 The three-band cancellation model

Noise Cancelling and DVNC divide the frequency spectrum; specifying the split stops either side over-claiming:

Layer	Band	Method
DVNC tonal	~20–250 Hz engine orders	AeroNode-fed synthetic-reference FxLMS — the differentiator
Noise Cancelling broadband	~50 Hz–1 kHz	H1 hybrid feedforward + feedback ANC (earcup mics)
Passive	~1 kHz and up	Earcup seal + materials

1.2 The seams every audio subsystem shares

- **One earcup driver + one in-cup error mic —** Noise Cancelling and DVNC both drive them; their two anti-noise outputs must sum into one (shared engine vs parallel-and-sum: open).
- **One H1 mic array —** Noise Cancelling and PureVoice share it; the in-earcup mic is both ANC feedback and a PureVoice reference. Geometry must be frozen jointly (with John).
- **One deterministic AeroNode↔H1 link —** CAN/RS-485-class wired or UWB/proprietary RF, **not standard Bluetooth.** DVNC's phase coherence and VGSA's head-tracked horizon both die on variable Bluetooth latency (20–100 ms). BT 5.x is comms/media only. See §6.
- **One set of ears —** comms, the VGSA horizon, DSMA cues, PilotSense warnings and media all contend. A single priority/mixing scheme must arbitrate them (§3.6).

1.3 The six working blocks

Block	Subsystem	Role in the stack
3.1	**Noise Cancelling**	Acoustic (mic-reference) hybrid ANC — the certifiable baseline & demo ANC pillar
3.2	**DVNC™**	Vibration-referenced ANC layer — the differentiator (off demo critical path)
3.3	**PureVoice™**	Voice capture + enhancement — Path B discriminative chain
3.4	**VGSA™**	Spatial-audio virtual horizon for disorientation (binaural, head-tracked)
3.5	**DSMA™ audio**	Stall-margin “headroom” cues — the audio-delivery + arbitration side

Block	Subsystem	Role in the stack
3.6	**Warnings & arbitration**	PilotSense audio alerts + the system-level audio priority/mixing scheme

3. Subsystem deep dives (the working blocks)

3.1 Noise Cancelling — acoustic hybrid ANC

What & why. The conventional acoustic (mic-reference) ANC path for the H1 earcup — feedforward (exterior) + feedback (in-cup) mics driving an in-cup canceller. It is the certifiable, self-contained baseline that works with the headset alone, and the ANC pillar for the December H1 Flight Demo. GA cockpit noise is 90–110 dB, LF-dominated; prop fundamental f0 = RPM × blade count ≈ 70–300 Hz. ANC owns 20–~1000 Hz; passive handles ≥1 kHz. Continuous LF processing is "acoustic fatigue" — cutting it is the human-performance payoff.

How. Committed architecture is hybrid AI-DSP: a real-time FxLMS (Filtered-x LMS) loop generates anti-noise (needs an accurate factory-identified secondary path Ŝ(z), driver→error-mic); a lightweight CRN (Convolutional Recurrent Network) supervises — it classifies the acoustic scene ("Cessna prop", "helicopter rotor", "jet wind") and retunes FxLMS step-size/coefficients, without sitting in the sample loop. Topology per the 10 Jul steer: hybrid feedforward+feedback + DVNC as a third reference. A fail-safe analog passthrough (comms works with the DSP powered down) is a non-negotiable V3 criterion.

Hardware to build it

- **Prototype:** RPi 4/5 (PREEMPT_RT) or Jetson
- **Production path:** STM32H7 / NXP i.MX RT + dedicated audio DSP (ADI ADAU1467)
- **CRN host:** AI-capable MCU — ARM Cortex-M55 + Ethos-U55 NPU
- **SigmaDSP bench:** EVAL-ADAU1787Z (earcup real-time path), EVAL-ADAU1467Z (algo dev), EVAL-ADUSB2EBZ USBi
- **Mics:** ≥3/earcup — reference + error + boom; low-latency MEMS (Knowles/Infineon), I²S/PDM
- **Fail-safe:** ADI ADG1634 audio switch vs relays; 6-pin Lemo cockpit connector + per-cockpit adapters; line-level conversion
- **Codecs:** ultra-low-latency ADC/DAC, round-trip <1 ms (non-negotiable); LF-capable driver

Software / algorithms

- **FxLMS / FxNLMS** adaptive core (public-domain — also the FTO defence)
- **CRN supervisor:** scene classifier; INT8 quantised; TFLite Micro / ONNX on the NPU
- **Ŝ(z) identification** procedure (factory white-noise)
- **SigmaStudio** for static hybrid ANC bring-up (adaptive FxLMS is NOT a drop-in block)
- **Training data:** aviation-noise corpora × clean speech (LibriSpeech), varied SNR; loss SI-SDR / log-mel
- **Deep Filter Net / RNNoise** evaluated for the comms path

When — status vs gate

WS-D · demo ANC pillar · next gate V2 (Month 4, ~1 Oct): acoustic FxLMS baseline running. On track — acoustic bench in progress (flight-audio testing underway); relay fail-safe passthrough bench test running now (due 12 Jul, result pending). Onboard ANC compute sizing kicked off this week (measure Deep Filter Net before locking CM5 + Hailo-8/Coral). FTO: Von Seidels ANR landscape Phase 1 underway (Zone 3 = the in-cup acoustic canceller — this is where the risk sits).

On the day — to align / decide

- **Silicon:** consumer ANC SoC (Qualcomm-class, closed canceller) vs owning the DSP — owning it keeps DVNC/PureVoice + FFT/HRTF load ours (John's FFT-accelerator-vs-NEON/NE10 question).
- **CRN scope:** scene classes, model size, NPU target, update rate.
- **Fail-safe switch location:** in the H1 (standalone-capable) vs in the AeroNode.
- **Test rig:** power + sample-rate budget; attenuation setup (ear simulator/HATS, DO-160G-class).

3.2 DVNC™ — vibration-referenced ANC

What & why. A vibration-referenced ANC layer (not a replacement) that uses the airframe-mounted AeroNode as a clean, predictive reference. The AeroNode extracts the engine/prop fundamental + harmonics through structure (IMU) and air (mic) and streams parameters (ω0, phase, per-harmonic amplitude — kilobits, not audio) to the H1, which synthesises matched anti-noise for the ~20–250 Hz tonal rumble. Defensibility is structural: no standalone headset (Bose A30, Lightspeed Delta Zulu) has an airframe-mounted reference node.

How — and the honest framing. H1 synthesises x(n) = [sin ω0n, cos ω0n, sin 2ω0n, …]; secondary path Ŝ(z) factory-fixed FIR; y(n)=W^T(n)x(n); W(n+1)=W(n) − μ·e(n)·x′(n), driven by the earcup error mic. Because vibration travels the airframe (~3,000–5,000 m/s) far faster than sound in air (~343 m/s), the AeroNode knows the engine state before the sound arrives. Do not overclaim: for a steady tone a phase-locked reference already cancels well — the defensible wins are (a) a clutter-free reference (deeper, faster convergence) and (b) tracking through RPM transients. P(z) (airframe→ear) is not static — needs continuous adaptation + dual-IMU head tracking.

Hardware to build it

- **Earcup ANC DSP:** EVAL-ADAU1787Z (headset-optimised, ~5 µs latency)
- **Algorithm-dev DSP:** EVAL-ADAU1467Z (MIPS + multichannel)
- **Programmer:** EVAL-ADUSB2EBZ (USBi)
- **Product-candidate silicon:** ADAU1860 / ADAU1797 (Lark Studio) — later
- **Mics/driver:** low-latency MEMS (avoid PDM decimation); LF-capable driver
- **AeroNode sensing:** calibrated IMU + reference mic (validated)
- **Bench (~$2,250 core):** miniDSP EARS + UMIK-2 + REW; LF monitor/sub; Scarlett 4i4 / MOTU M4; Saleae Logic 8; bench PSU
- **Phase 2 (deferred, ~$9k):** HATS / IEC 60318-4 ear simulator

Software / algorithms

- **Order tracking:** PLL / FFT on IMU → ω0, phase, per-harmonic amplitude
- **Synthetic-reference FxLMS** (multi-reference adaptive notch)
- **Ŝ(z) offline identification;** P(z) online adaptation + dual-IMU head-track phase correction
- **Offline proof:** Python (numpy/scipy/sounddevice) on recorded IMU+mic — per-harmonic dB reduction (the V2 evidence)
- **SigmaStudio / REW** for hardware bring-up + measurement
- **Stage-3 (layered, optional):** GFANC — CNN picks the control filter per scene; fast loop applies it

When — status vs gate

WS-D differentiator · OFF the demo critical path · reference engine demonstrable by V2, matures Months 6–12. Concept validated (AeroNode IMU+mic engine signature captured on the airframe). Next: offline Python proof on the recorded signatures. Bench work waits on dev-kit delivery. IP: DVNC is the filing focus but **not yet filed** — keep disclosure high-level.

On the day — to align / decide

- **Link:** wired CAN/RS-485 vs UWB/proprietary RF — determinism + latency + clock/phase-sync budget (the requirement that makes DVNC work).
- **Accelerometer selection/placement** + mounting/vibration isolation for a clean structural reference (joint with AeroNode).
- **P(z) strategy:** pre-trained per airframe vs continuous online; validity across aircraft types.
- **NC↔DVNC summation** at the shared earcup: one engine vs parallel-and-sum.

3.3 PureVoice™ — voice capture & enhancement

What & why. The pilot-voice capture + enhancement chain (into comms, AeroVault recording, Creator Mode). The governing decision is already made: Path B — discriminative enhancement — is the committed primary path; generative resynthesis is retained only as an extreme-SNR fallback. Principle: "preserve and attenuate; never fabricate." Worst failure is over-suppression (muffled/dropped consonant) — recoverable, and never a substitution. It fails toward noisy-but-correct, not clean-but-wrong. Generative resynthesis can confidently synthesise a wrong word ("descend" for "ascend", a corrupted altitude digit) — so it is out of the safety path. Certifiability was the deciding factor.

How. Three decoupled problems. Capture: multi-sensor fusion — NC boom mic (air, primary) + in-earcup reference mic (doubles as ANC ref) + bone-conduction sensor (temple/concha piezo). Clean: 2-mic adaptive Griffiths-Jim cancellation → compact mask-based neural denoiser (bounded gain). Encode: low-delay codec (Opus-LD / LC3plus). Two latency budgets: the sidetone loop (own-voice monitor) is hard — DAF disruption climbs ~50→200 ms, so the monitor never includes the codec/radio; the transmit path (~20–30 ms) carries the heavy enhancement.

Hardware to build it

- **Boom mic:** noise-cancelling, air (primary capture)
- **In-earcup reference mic:** shared with ANC feedback
- **Bone-conduction sensor:** temple/concha piezo (voicing pickup) — selection open
- **Compute location:** headset DSP vs AeroNode (NPU/TPU) — open budget question
- **Shares the H1 mic array** with Noise Cancelling (geometry freeze pending)

Software / algorithms

- **Griffiths-Jim** 2-mic adaptive cancellation
- **Mask-based neural denoiser** (bounded gain on the observed spectrum)
- **Codec:** Opus-LD vs LC3plus (per link/use-case)
- **Sidetone:** minimally-processed / analog low-latency monitor path
- **Fallback (contained):** LPC + LPCNet-hybrid vocoder, extreme-SNR only
- **RV-1 verification:** reconstruction-integrity evidence before sole-live-ATC use

When — status vs gate

WS-D (D-Voice) · off the demo critical path · gate RV-1. Committed to Path B. RV-1 evidence work = the M3–4 neural-vocoder bench investigation (due ~1 Oct). Mic-array geometry freeze rising in urgency now that headset ID CAD has started.

On the day — to align / decide

- **Joint mic architecture** with Noise Cancelling + H1 mechanical: placement, sampling clocks, which stage owns which mics — freeze before either side commits.
- **Denoiser model + compute location** (headset vs AeroNode).
- **Codec** (Opus-LD vs LC3plus) and **bone-conduction sensor + fusion weighting.**

3.4 VGSA™ — spatial-audio virtual horizon

What & why. Experimental, emergency-only assistance for spatial disorientation in IMC. The AeroNode's high-rate IMU drives a 3D binaural "virtual horizon" that wraps the pilot's head and shifts with pitch/roll/yaw, prompting instinctive corrective inputs — a non-visual auditory horizon that persists when the visual one is gone. Spatial disorientation holds ~15% of fatal mishaps.

How. Localisation from ITD (<1500 Hz, azimuth), IID (>1500 Hz, azimuth) and spectral/pinna filtering (>4 kHz, elevation); binaural cues alone can't resolve elevation or front/back, so the HRTF does. HRTFs update in real time with head tracking so the source stays locked to the aircraft frame. Attitude from the AeroNode IMU; head orientation from the H1's own IMU (relative to aircraft); real-time HRTF convolution → binaural on the H1 drivers. Latency is safety-critical — the corrective response breaks if head-motion-to-audio lag is too high (hence the wired link, <1 ms).

Hardware to build it

- **Attitude:** AeroNode high-rate IMU (locked to airframe via the structural dock)
- **Head tracking:** H1 IMU (orientation relative to aircraft)
- **Output:** H1 binaural drivers
- **Dev kits:** Pixhawk 6X, 2× CM5, IMUs (ordered 2 Jul; arrival unblocks the bench tone)
- **Deterministic wired link** mandatory (latency-critical)

Software / algorithms

- **Real-time HRTF convolution** (generic vs individualised — open)
- **Head-track fusion:** H1 IMU vs AeroNode IMU → relative orientation per earcup
- **Front/back-ambiguity mitigation**
- **V1 bench:** minimal IMU-driven spatial tone
- **Validation:** blindfold attitude-correction protocol (→ V3 gate)

When — status vs gate

WS-C · H1-only · next gate V3 (Month 6, ~1 Dec): blindfold VGSA validated (attitude correction on audio alone). Near-term V1 item (~1 Aug): IMU-driven tone on the bench once dev kits land. Standing #1 open item: end-to-end latency budget + measurement method.

On the day — to align / decide

- **End-to-end latency budget** (IMU → AeroNode → H1 drivers) + measurement method — the standing #1.
- **Head-tracking approach** (headset IMU vs AeroNode IMU fusion); HRTF generic vs individualised.

3.5 DSMA™ — audio delivery of stall-margin cues

What & why (audio scope). DSMA estimates synthetic airspeed (EKF on IMU+GPS+baro), computes the dynamic stall speed from G-load [V_stall(dyn) = V_stall(base) × √n], and delivers a safety-margin "headroom" cue through the headset — not a binary stall horn. States: SAFE (silent) → CAUTION (soft "Airspeed decaying") → EMERGENCY (priority "WARNING: STALL MARGIN CRITICAL"). For John the relevant part is the audio delivery: how the headroom is encoded to sound, injected into the pilot's stream, and prioritised against everything else.

How (audio path). In the PoC bench a dedicated audio DSP (ADAU1701-class) mixes TTS warning triggers (sent from the Pi over analog line-in) into the ANC audio stream; flap state uses a Picovoice Porcupine wake-word engine on the intercom (offloaded to a Coral Edge TPU so audio has no latency). In production this becomes the H1 audio path + the audio-arbitration contract.

Hardware to build it

- **Flight controller:** Pixhawk 6X (ArduPilot / EKF3)
- **Companion + edge AI:** RPi 5 (8 GB) + Google Coral Edge TPU (wake-word offload)
- **Audio DSP:** ADAU1701-class eval board (TTS injection + mix into ANC stream)
- **Link:** Ethernet (MAVLink/UDP) primary; UART fallback

Software / algorithms

- **stall_monitor.py** (50 Hz PyMAVLink loop) — margin + state machine
- **Picovoice Porcupine** wake-word ("Flaps ten/ extended") + kinematic flap detection
- **Cue encoding:** headroom → sound (design open)
- **TTS trigger** over GPIO/line-out on threshold breach
- **Turbulence cross-check** (n ≈ 1/cos bank) to suppress false alerts

When — status vs gate

WS-B · next gate V2 (Month 4): SAFE/CAUTION states firing from replayed flights; rides ArduPilot/EKF3. V1 (~1 Aug) owes the stall_monitor.py core loop in SITL. AeroNode Lite ships DSMA post-flight only; live audio cues need the full-stack AeroNode/H1 path.

On the day — to align / decide

- **Audio cue design** (headroom encoding) and its **priority in the arbitration scheme** (§3.6).

3.6 Warnings & the audio-arbitration layer

What & why. PilotSense biometrics (HR, HRV, SpO2, CO2 via in-ear + temple sensors; Bio-Audio Fusion engine) deliver prioritised audio warnings for fatigue, stress, hypoxia and hypercapnia. Together with comms, the VGSA horizon, DSMA

cues and media, that is five audio sources contending for two ears. A single priority + mixing scheme must own what ducks what, what is spatialised, and what is never suppressed (safety-critical audio: stall horn, traffic/terrain, ATC). This is a root-level system contract — no single subsystem may define it — and a natural home for DSP leadership.

Why this is the keystone block

Every other block outputs into this one. The arbitration scheme constrains latency budgets (VGSA, sidetone), mixing headroom (DVNC + Noise Cancelling anti-noise + cues), and safety guarantees (nothing critical is ducked). Getting it specified early de-risks all five.

On the day — to align / decide

- **Priority order + ducking rules** across comms / VGSA / DSMA / PilotSense / media.
- **What is spatialised** vs mono; what is never suppressed.
- **Where arbitration runs** (AeroNode vs H1) and its latency budget.

4. Consolidated hardware to build the audio stack

Everything the audio blocks need, grouped. Shared items appear once; per-block specifics are in §3.

Group	Item	Used by / note
ANC DSP (real-time)	ADI EVAL-ADAU1787Z	NC + DVNC earcup path (~5 µs latency)
Algorithm-dev DSP	ADI EVAL-ADAU1467Z	NC + DVNC FxLMS + fusion dev
DSP programmer	EVAL-ADUSB2EBZ (USBi)	SigmaStudio interface
Product-candidate silicon	ADAU1860 / ADAU1797 (Lark Studio)	Later-phase NC codec
CRN / NPU host	Cortex-M55 + Ethos-U55 (or CM5 + Hailo-8 / Coral)	NC scene classifier; compute sizing open
Flight controller	Pixhawk 6X (ArduPilot/EKF3)	DSMA + VGSA attitude/dynamics
Companion compute	RPi 5 (8 GB) / CM5	DSMA logic, edge AI, DVNC order tracking
Edge TPU	Google Coral	DSMA wake-word offload
Mics	Low-latency MEMS ×≥3/earcup + boom + bone-conduction piezo	NC (ref/error), PureVoice (boom/BC), shared array
Driver	LF-capable transducer	All cancellation is sub-1 kHz
Fail-safe / interconnect	ADG1634 switch (vs relays), 6-pin Lemo, per-cockpit adapters	NC passthrough; AeroNode plug terminal
AeroNode↔H1 link	CAN/RS-485-class wired or UWB/RF	DVNC + VGSA determinism (not Bluetooth)
Measurement (Phase 1, ~$2,250)	miniDSP EARS + UMIK-2 + REW; LF monitor/sub; Scarlett 4i4 / MOTU M4; Saleae Logic 8; bench PSU	ANC-depth + bus debug
Measurement (Phase 2, ~$9k)	HATS / IEC 60318-4 ear simulator	Formal eardrum-reference validation — deferred

5. Consolidated software & algorithm stack

Layer	Tools / methods
Real-time ANC	FxLMS / FxNLMS adaptive core; Ŝ(z) factory identification; synthetic-reference FxLMS (DVNC); SigmaStudio for static-ANC bring-up
AI / neural	CRN scene classifier (INT8, TFLite Micro / ONNX); GFANC (Stage 3); mask-based neural denoiser (PureVoice); Deep Filter Net / RNNoise (comms path)
Voice	Griffiths-Jim adaptive cancellation; Opus-LD / LC3plus codecs; LPCNet-hybrid fallback
Spatial	Real-time HRTF convolution; IMU head-track fusion
Flight / cues	ArduPilot (EKF3); stall_monitor.py (PyMAVLink 50 Hz); Picovoice Porcupine wake-word; MAVLink over UDP
Offline / bench	Python DSP (numpy / scipy / sounddevice) — primary free proof path; REW; SigmaStudio (Windows)
Prototyping OS	RPi PREEMPT_RT; Goose (internal Qwen-based knowledge/LLM assist)

6. The deterministic AeroNode↔H1 link & fail-safe

This one interface enables DVNC and VGSA and gates the fail-safe story, so it deserves its own block. It must carry DVNC parameters (kilobits — ω0/phase/harmonics) and VGSA/DSMA/PilotSense audio with deterministic, low-latency, phase-coherent timing. Standard Bluetooth (20–100 ms, variable) is disqualified for the real-time path; BT 5.x stays for comms/media/AeroSync. Candidates: wired CAN/RS-485-class (front-runner in the DEV view — clock/phase sync is the real requirement, not bandwidth) or UWB/proprietary RF. The same cable/connector carries the fail-safe analog passthrough — which is why the switch-location decision (H1 vs AeroNode) is shared between Noise Cancelling and this link.

On the day — to align / decide

- **Bus vs RF** + determinism/latency/clock-sync budget.
- **Fail-safe switch location** (H1 standalone vs AeroNode-as-unit) and its effect on the plug terminal + per-cockpit adapters.

7. Running order — Wednesday 15 July

Attendees: John, Chris, Peter, Duran. Duran already answered John's technical-questions list (10 Jul) with three steers: topology is hybrid + DVNC third reference; AeroNode→H1 is a deterministic wired link, not Bluetooth; the silicon question is consumer ANC SoC vs owning the DSP. Each block below maps to a section above.

Time	Block	Aim
10 min	Architecture (§1)	Shared model: AeroNode split, three-band cancellation, the four seams
25 min	Noise Cancelling (§3.1)	Silicon vs FTO Zone 3; CRN scope; fail-safe (relay bench result); test rig
20 min	DVNC (§3.2)	Synthetic-reference FxLMS; the link; $\hat{S}(z)/P(z)$; offline proof plan
15 min	PureVoice (§3.3)	Path B; joint mic architecture; latency budgets; codec/denoiser/BC
15 min	VGSA + DSMA (§3.4–3.5)	HRTF + head-track + latency; cue encoding
15 min	Arbitration + link (§3.6, §6)	Priority/mixing scheme; wired-link + fail-safe switch location
10 min	Workplan & contract	3-day cadence; first deliverables; Sigma Studio baseline; wrap contractor structure

Suggested first deliverables for John (to frame the workplan)

1. Offline Python DVNC proof — per-harmonic dB reduction on the recorded AeroNode IMU+mic signatures (the V2 evidence).

2. $\hat{S}(z)$ factory-identification procedure (shared by Noise Cancelling + DVNC).

3. A first draft of the audio-arbitration priority/mixing scheme (unblocks all five blocks).

Until the contract is signed, ~90% of initial ANC/DVNC work is doable in-house on the Sigma Studio baseline.

8. Safety & IP notes

- **Safety framing (mandatory).** VGSA, DSMA and the PilotSense alerts are experimental, advisory, emergency-only aids — they augment pilot awareness and do not replace certified avionics or training. The AeroNode has zero input to flight controls. Noise Cancelling/DVNC are comfort features; the fail-safe analog passthrough guarantees comms with the DSP powered down, and the H1 always works as a passive aviation audio headset.
- **FTO.** Risk concentrates in the in-cup acoustic canceller (Noise Cancelling, Zone 3). The FxLMS/FxNLMS + open-source-model route is the independent-derivation defence. Von Seidels ANR landscape Phase 1 is underway; check any in-cup design choice against Zone 3 before committing silicon.
- **DVNC disclosure.** DVNC is the differentiator and filing focus but is NOT yet among the filed provisionals — keep external disclosure high-level until filed. Treat this document as confidential IP under development.

Appendix — Glossary

Term	Meaning
FxLMS / FxNLMS	Filtered-x (Normalised) LMS — the adaptive filter at the heart of both ANC paths; public domain since the 1980s
Ŝ(z) / P(z)	Secondary path (driver→error-mic, factory-fixed) / primary path (airframe or noise→ear, non-static)
CRN	Convolutional Recurrent Network — ANC scene classifier supervising FxLMS (not in the sample loop)
GFANC	Generative Fixed-filter ANC — a CNN selects a control filter per scene; a fast loop applies it
HRTF	Head-Related Transfer Function — spectral mapping that places a binaural source in 3D (VGSA)
ITD / IID	Interaural Time / Intensity Difference — azimuth localisation cues
Griffiths-Jim	Adaptive 2-mic beamformer/canceller used in the PureVoice clean stage
Path B	PureVoice's committed discriminative chain (“preserve & attenuate, never fabricate”)
DAF	Delayed Auditory Feedback — the sidetone latency constraint (~50–200 ms)
RV-1	PureVoice reconstruction-integrity gate before sole-live-ATC use
EKF3	ArduPilot's Extended Kalman Filter state estimator — DSMA/VGSA source
DO-160G	RTCA environmental test standard referenced for attenuation/qualification

Source docs (Drive · Technologies/): DEV Noise Cancelling.md, DEV DVNC.md, DEV PureVoice.md, DEV VGSA.md, DEV DSMA.md + root interface-contract CLAUDE.md; timing/status from _AeroCore/ Aerosense Flight Plan and Preflight 2026-07-08.

Aerosense Aviation (Pty) Ltd — internal / confidential.